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Machine learning in stock market forecasting: a comprehensive review

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Abstract

Forecasting financial markets remains difficult because price series are noisy, nonlinear, nonstationary, and subject to structural breaks. This structured review synthesizes empirical machine learning (ML) and deep learning (DL) research on equities, indices, commodities, foreign exchange, and cryptocurrencies, covering classical ML, recurrent networks, CNN-based models, attention and Transformer architectures, multimodal and graph-based systems, and reinforcement-learning approaches. Because studies differ in task formulation, horizon, data frequency, target variable, feature set, validation design, and metric choice, the review combines qualitative synthesis with structured quantitative evidence mapping rather than a single pooled effect size. The paired-error aggregation identified 17 peer-reviewed studies and 47 same-dataset, same-horizon proposed-versus-baseline error comparisons. On a study-level median basis, this subset showed a median relative error reduction of 20.3%, with an interquartile range of 5.7%-50.7% and a full range of −0.8%-71.5%. These results indicate that ML/DL methods often improve reported performance in controlled experiments, but they do not establish universal real-world superiority. Comparability remains limited by validation design, transaction costs, slippage, market regimes, and inconsistent uncertainty reporting. Future work should prioritize leakage-safe validation, calibration, interpretability, uncertainty quantification, and realistic economic evaluation.

Keywords Machine learning, Deep learning, Stock market forecasting, Financial time series, Transformers, Sentiment analysis, Explainable AI

JEL classification C45, C52, C53, C58, G17

1 Introduction

Forecasting the stock market remains a difficult problem for financial professionals and researchers because financial prices are volatile, nonlinear, nonstationary, and sensitive to changing macroeconomic and geopolitical conditions [1–3]. Traditional statistical approaches, including Generalized Autoregressive Conditional Heteroscedasticity (GARCH) and Autoregressive Integrated Moving Average (ARIMA) models, often rely on assumptions of linearity, stationarity, or stable error structure. These assumptions can weaken under dynamic market conditions, limiting the practical effectiveness of purely econometric approaches in some settings [4]. In response, machine learning (ML)

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and deep learning (DL) methods have become useful tools for modeling nonlinear relationships, temporal dependencies, and high-dimensional financial inputs [2].

Despite significant advances from early Support Vector Machines [5] and Random Forest implementations [6] to standalone architectures such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs), several gaps remain. Many earlier reviews do not fully address recent Transformer variants, graph-based models, time-series foundation models, or domain-adapted language models such as FinBERT. In addition, interpretability, uncertainty quantification, and multimodal pipelines that integrate price data with news, macroeconomic indicators, and cryptocurrency data remain unevenly covered. These gaps motivate a review that emphasizes not only model evolution but also the evidentiary strength and robustness of reported forecasting gains.

Because this article is a structured literature review rather than a primary empirical experiment, it does not test a formal causal hypothesis. Instead, it is guided by three research questions:

RQ1: How have financial forecasting methods evolved from classical econometric and early machine learning approaches toward deep learning, attention-based, multimodal, and reinforcement-learning systems?

RQ2: Under what experimental conditions do ML/DL models report measurable improvements over traditional or study-specific baselines?

RQ3: What limitations in validation design, robustness testing, uncertainty reporting, and economic evaluation constrain claims about real-world forecasting performance?
Accordingly, the objective of this review is not only to summarize model families, but also to evaluate whether reported forecasting gains are supported by extractable quantitative evidence and whether those gains remain credible under realistic financial-market constraints.

1.1 Significance and novelty

This review differs from earlier surveys by focusing on the gap between reported model accuracy and deployable forecasting reliability. Rather than treating deep learning performance claims as uniformly comparable, the review separates model families, market domains, feature regimes, and evaluation settings so that improvements are interpreted in context. It also gives particular attention to recent transformer, foundation-model, multimodal, graph-based, and sentiment-aware approaches that are increasingly common in post-2020 financial forecasting research.

Although prior reviews often catalogue model families and report headline accuracy gains, many do not clearly separate reported experimental performance from deployable forecasting reliability. This distinction is central in financial forecasting because studies frequently differ in target construction, prediction horizon, data frequency, baseline strength, validation design, and treatment of market frictions. The present review therefore treats financial forecasting as both a modeling problem and an evidence-evaluation problem, emphasizing whether reported improvements remain credible under leakage-safe validation, regime shifts, uncertainty reporting, and economic-validity constraints.

A second contribution is the structured aggregation of extractable model-versus-baseline error comparisons. Because the included studies differ in target variables, horizons, data frequency, scaling conventions, and metrics, the review avoids a single pooled effect size and instead reports study-level summaries for directly comparable paired-error evidence. Seventeen peer-reviewed studies supplied 47 same-dataset, same-horizon proposed-versus-baseline error comparisons; directional, trading-oriented, robustness, and uncertainty evidence is mapped separately.

1.2 Motivation and target audience

Deep learning methods for financial prediction are evolving quickly, especially through Transformers and multimodal data fusion. At the same time, many papers report strong results without aligning evaluation choices or accounting for regime shifts, costs, and data limitations. This review provides a structured reference covering established methods and current practice, with attention to robustness and uncertainty where reported.

This review is intended for researchers studying financial machine learning, engineers and data scientists building forecasting systems, and finance professionals who want a clear picture of model inputs, outputs, and reliability limits. It is also designed to be accessible to students and newcomers who need a structured starting point.

2 Methodology

2.1 Review design and reporting standard

This article was designed as a structured literature review to maximize transparency and reproducibility. The workflow follows PRISMA-style expectations for reporting systematic reviews, including an explicit search strategy, eligibility criteria, screening procedure, and a study-selection flow description.

2.2 Information sources (databases)

To ensure broad coverage across finance, econometrics, and machine learning venues, studies were retrieved from a combination of multidisciplinary indexing databases and field-specific libraries. Core databases included Scopus, Web of Science, and IEEE Xplore, complemented by targeted searches in ACM Digital Library, ScienceDirect, SpringerLink, and Google Scholar to capture relevant conference proceedings and early-access articles that are common in machine learning research.

2.3 Search strategy and query construction

Searches were executed using keyword strings composed of three concept blocks:

(1) financial forecasting domain (e.g., "stock market", "equity", "index", "financial time series", "volatility", "returns", "cryptocurrency"),
(2) model family (e.g., "machine learning", "deep learning", "neural network", "LSTM", "GRU", "CNN", "Transformer", "attention"), and
(3) exogenous signals/multimodal inputs (e.g., "sentiment", "news", "social media", "macroeconomic", "multimodal", "fundamentals", "order book").

A representative boolean query (adapted per database syntax) was:

("stock market" OR equity OR "stock index" OR cryptocurrency OR "financial time series") AND

(forecast OR predict* OR "price direction" OR volatility OR return*)*

AND ("machine learning" OR "deep learning" OR LSTM OR GRU OR CNN OR Transformer OR attention OR "reinforcement learning")

AND (sentiment OR news OR "social media" OR multimodal OR macroeconomic OR "order book").

Filters were applied to prioritize empirical work (i.e., studies reporting evaluation metrics) and to retain studies that either (a) introduced a modeling approach or (b) presented a comparative evaluation of established methods. Because state-of-the-art ML results often appear first in conference proceedings and preprints, preprints were retained only when they were clear methodological contributions or widely used baselines in subsequent peer-reviewed research; these were flagged during extraction.

The initial record count should therefore be interpreted in light of these constraints: retrieval was limited to English-language empirical or methodologically relevant studies, database-specific search filters were applied where available, and finance-NLP papers without a forecasting outcome were removed during screening. This constrained search design explains why a broad conceptual query produced a smaller evidence base after de-duplication and eligibility screening.

2.4 Time window and targeted search

To reflect the evolution from classical machine learning to modern deep learning and Transformer-based approaches, the review targeted literature from 2014 onward, with emphasis on post-2018 deep learning growth and post-2020 Transformer/time-series attention models. The main search was completed in April 2026. A targeted search identified empirical studies, including recent 2024–2025 work, that reported directly extractable proposed-model and baseline error values on the same dataset and forecast horizon. Retrieval was constrained to English-language empirical studies, conference papers, peer-reviewed articles, and methodologically relevant preprints that reported usable forecasting or trading metrics. Google Scholar searches were used as a supplementary source rather than as the sole evidence base, and candidate records from Google Scholar were screened against the same inclusion and exclusion criteria as database records.

2.5 Eligibility criteria (inclusion and exclusion)

Studies were included if they met all of the following criteria:

- Topic relevance: addressed forecasting or prediction tasks in financial markets (stocks, indices, commodities, FX, or cryptocurrencies).
- Method relevance: employed ML/DL methods (e.g., tree ensembles, SVM, RNN/LSTM/GRU, CNN, Transformer/attention architectures, hybrid/ensemble models, or reinforcement-learning-based trading/forecasting pipelines).
- Empirical evaluation: reported at least one quantitative performance metric (e.g., RMSE, MAE, MAPE, directional accuracy, F1/accuracy for classification, Sharpe ratio for trading-based evaluation).
- Sufficient reporting: described inputs, target variable(s), prediction horizon, and evaluation setup clearly enough to support synthesis.

Studies were excluded if they:

- were purely conceptual/opinion pieces without empirical results;
- did not focus on financial time-series prediction (e.g., finance NLP tasks not linked to forecasting outcomes);
- lacked sufficient methodological detail (e.g., missing data description or evaluation procedure);
- were duplicates or near-duplicates of the same experiment across venues (the most complete version was retained).

2.6 Study selection and screening workflow (PRISMA-style) selection followed a staged screening process

(1) Identification: database searches were executed and records exported to a reference manager.
(2) De-duplication: duplicate and near-duplicate records across databases were removed; 412 records remained after de-duplication.
(3) Title/abstract screening: 412 records were screened for relevance to financial forecasting and ML/DL methods; 264 records were excluded at this stage for being out of scope, non-empirical, or insufficiently connected to forecasting outcomes.
(4) Full-text eligibility: 148 full texts were assessed against inclusion/exclusion criteria; 45 were excluded for reasons such as no empirical metrics, NLP-only analysis without forecasting outcomes, insufficient methodological detail, or duplicate experimental reporting.
(5) Final inclusion: 103 studies were included in the qualitative synthesis. A 38-study subset was coded for structured quantitative evidence mapping, and directly extractable paired comparisons were used for conservative numerical aggregation when available.

The systematic search and screening process, conducted in accordance with PRISMA-style reporting expectations, resulted in 777 total records, 412 records after de-duplication, 148 full texts assessed for eligibility, 103 studies included in qualitative synthesis, and 38 studies coded for structured quantitative evidence mapping. Within this evidence map, a targeted paired-error search identified peer-reviewed studies with extractable same-dataset, same-horizon proposed-versus-baseline error values. The final direct paired-error aggregation included 17 peer-reviewed studies and 47 paired comparisons. The step-by-step filtering process and nested evidence levels are shown in Fig. 1.

2.7 Data extraction and coding schema

A standardized extraction template was used to ensure consistent synthesis across heterogeneous studies. For each included paper, the following fields were recorded:

- Market domain: stocks/indices/commodities/FX/cryptocurrency.
- Asset granularity and frequency: single-asset vs multi-asset; daily vs intraday vs tick/order book.
- Prediction task: regression (price/return/volatility) vs classification (direction/movement) vs trading policy optimization.
- Model class: classical ML, RNN-family, CNN-family, Transformer-family, hybrid/ensemble, multimodal, RL.

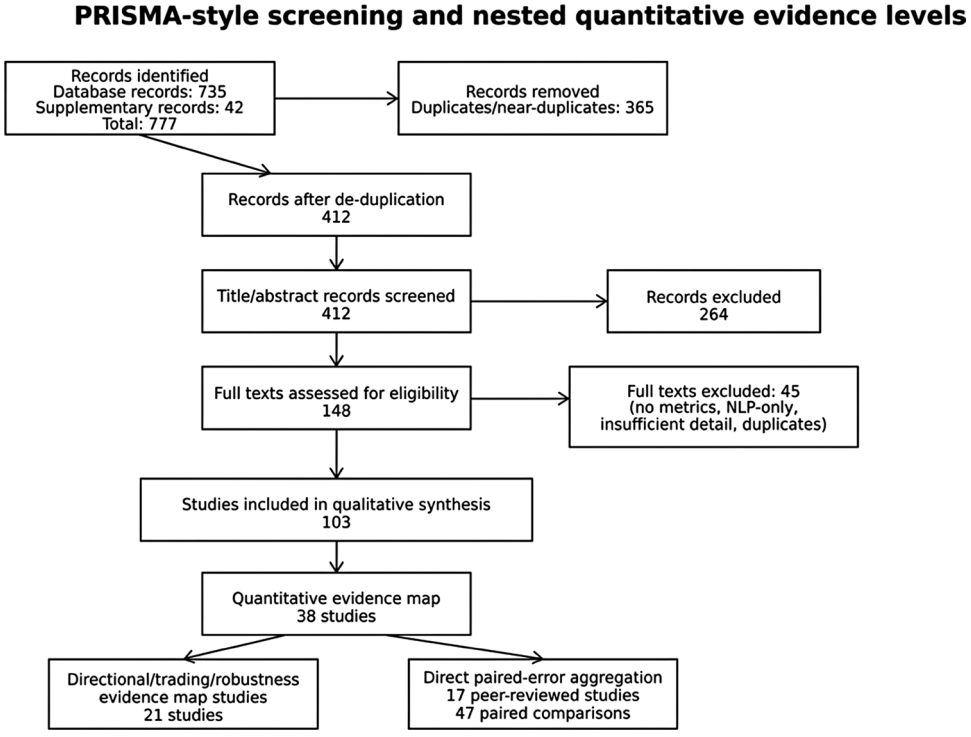


Fig. 1 PRISMA-style study selection and nested quantitative evidence levels. The review identified 777 records, retained 412 after de-duplication, assessed 148 full texts, included 103 studies in qualitative synthesis, and coded 38 studies in the structured quantitative evidence map. A targeted paired-error search produced a direct aggregation of 17 peer-reviewed studies and 47 same-dataset, same-horizon paired comparisons

- Inputs/features: OHLCV, technical indicators, fundamentals, macroeconomic variables, text/news/social sentiment, order book features.
- Horizon and evaluation protocol: train/validation/test split logic, rolling window vs static split, and leakage safeguards.
- Metrics: RMSE/MAE/MAPE, directional accuracy, classification metrics, and (when applicable) risk-adjusted returns such as Sharpe ratio.
- Key findings: main performance claims, ablations, and reported limitations.

To improve comparability, extracted results were annotated with metric definitions and scaling context (e.g., whether metrics were computed on normalized prices, returns, or de-normalized prices) and flagged when direct comparison was not meaningful.

Title and abstract screening and initial data extraction were performed by one reviewer. Ambiguous eligibility decisions and extracted quantitative values were subsequently checked by a second author, and disagreements were resolved through discussion and consensus.

2.8 Quality appraisal (risk of bias and robustness checks)

Given the fragility of many financial forecasting claims, each study was assessed using a practical robustness checklist:

- Leakage control: whether feature construction and scaling were performed strictly within training folds/windows.
- Evaluation realism: use of time-ordered splits and rolling/forward validation rather than random splits.

- Baseline strength: inclusion of meaningful baselines (e.g., ARIMA/GARCH, tree ensembles, common DL baselines).
- Replicability: availability of dataset description, hyperparameters, and training procedure.
- Sensitivity/ablation reporting: support via ablations or sensitivity analyses.
- Economic validity (for trading papers): inclusion of transaction costs, slippage, and realistic execution assumptions.

Studies were not excluded solely for weaker reporting; instead, quality flags were used to interpret findings conservatively in the synthesis.

2.9 Evidence synthesis and quantitative aggregation

Evidence synthesis was conducted in two layers. First, all eligible studies were grouped qualitatively by market domain, task type, model family, feature scope, and evaluation design. Second, a structured quantitative evidence map was developed for studies reporting extractable forecasting, classification, trading, robustness, or uncertainty outcomes. The paired-error component focused specifically on studies reporting proposed-model and baseline RMSE, MSE, MAE, or MAPE values on the same dataset and forecast horizon.

Because the included studies differed substantially in asset class, prediction horizon, data frequency, target variable, feature set, scaling procedure, and evaluation metric, a single pooled meta-analytic effect size was not computed. Instead, results were stratified by task and metric type. For error-based studies reporting RMSE, MSE, MAE, or MAPE, improvement was calculated as relative error reduction compared with the strongest relevant external baseline reported within the same study: Relative error reduction (%) = [(baseline error - proposed model error)/baseline error] x 100.

For studies reporting multiple assets, horizons, or experimental settings, each same-dataset, same-horizon comparison was first extracted as a paired comparison and then summarized as a study-level median.

Cross-study aggregation was then performed using these study-level medians so that studies with many assets or horizons did not dominate the synthesis. Directional forecasting studies, trading-oriented studies, and uncertainty/robustness evidence were summarized separately because accuracy, F1, Sharpe ratio, annualized return, calibration, and interval coverage are not commensurable with point-forecast error metrics.

Aggregated paired-error results are therefore reported using counts, study-level medians, interquartile ranges, and full ranges rather than a single pooled effect estimate. This approach allows conservative quantitative comparison while avoiding misleading precision across heterogeneous financial forecasting tasks. The interquartile range for the paired-error aggregation was computed from the 17 study-level medians using an exclusive median-of-halves convention, excluding the overall median before calculating Q1 and Q3.

2.10 Reproducibility statement

Search strings, screening logic, extraction fields, quantitative coding rules, and paired-error aggregation rules are summarized in Sects. 2.3-2.9 and Appendix B to support repeatability. The paired-error subset was separated from directional, trading-oriented, and supporting quantitative evidence. Appendix B identifies each eligible study, the

number of extracted pairs, the baseline choice, the study-level median relative error reduction, and any caveats affecting interpretation.

3 Forecasting background

3.1 Characteristics of financial time series

Financial time series (prices, returns, volatility) are difficult to forecast because they exhibit several well-documented properties: **nonstationarity**, **volatility clustering**, **seasonality**, **cross-asset interdependencies**, **leverage effects**, and **structural breaks** [7–12]. These characteristics violate assumptions behind many classical forecasting methods and motivate flexible models that can adapt to regime changes and nonlinear dynamics (Fig. 2).

3.2 Forecasting setup: inputs, horizons, and uncertainty

Forecasting can be viewed as learning a mapping from historical market information and optional exogenous signals to a future target at horizon h. Formally, many forecasting studies can be summarized as estimating $y_{t+h} = f(X_{t-L:t}, Z_{t-L:t}) + \varepsilon_{t+h}$, where X denotes historical market variables such as OHLCV, Z denotes optional exogenous inputs such as technical indicators, macroeconomic variables, order-book states, or sentiment embeddings, L denotes the look-back window, h denotes the prediction horizon, and ε represents unexplained forecast variation. The residual term is used here as a general theoretical representation of forecasting error; most reviewed studies do not explicitly model it as a separate stochastic process and instead handle unexplained variation implicitly through losses such as MSE, MAE, cross-entropy, or trading-objective functions. Horizons are handled either recursively, by iterating one-step predictions, or directly, by learning one or more future horizons at once. Forecasts are most often reported as point estimates, while prediction intervals, calibration, and interval coverage are reported less consistently.

3.3 General workflow of forecasting operations

Most stock market prediction studies follow a broadly standardized pipeline [3, 13]. A complete workflow begins with problem definition, including the target variable, forecast horizon, asset universe, and intended use case. After this step, studies typically

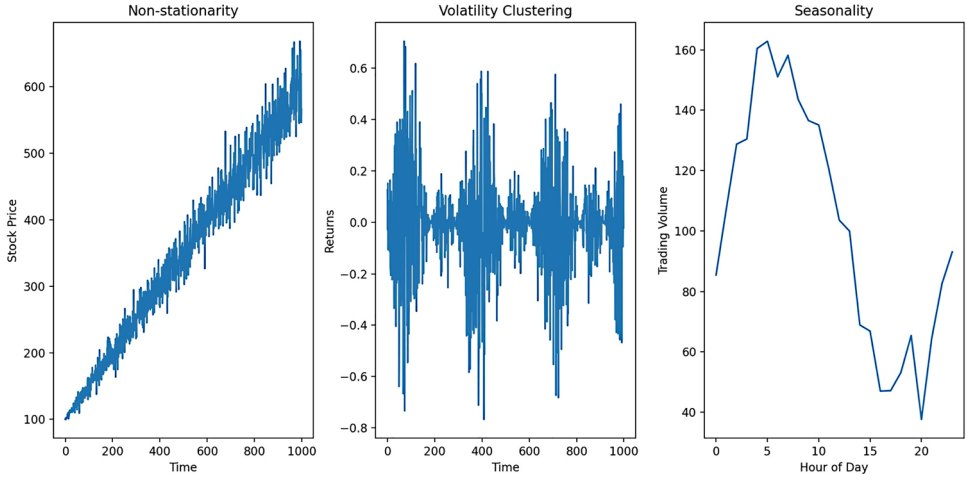


Fig. 2 Common time series Characteristics

proceed to data collection, using OHLC prices and volume from sources such as Yahoo Finance, Bloomberg, or Tushare, with evaluation commonly framed around equity indices, individual securities, or market-specific benchmark datasets rather than a single standardized asset universe.

Next, **preprocessing** converts raw market data into a usable form by handling missing values, applying normalization, and reducing noise using methods such as wavelet transforms or Empirical Mode Decomposition (EMD). Some studies also incorporate visual transformations (e.g., Heiken-Ashi candlesticks) to support trend representation [14].

After preprocessing, feature engineering constructs predictors intended to capture market dynamics, including technical indicators (e.g., RSI, MACD), macroeconomic variables, and sentiment signals derived from financial news or social media. Combining these feature types has been shown to improve prediction accuracy in many settings, especially when numerical and textual signals are aligned without look-ahead leakage [15, 16].

Researchers then move to model selection, choosing among machine learning and deep learning architectures such as RNNs, CNNs, Transformers, and hybrid systems based on the data frequency, horizon, and forecasting task [2, 13]. Training and validation are conducted using time-ordered splits to tune hyperparameters and assess generalization, and performance evaluation is typically reported using metrics such as RMSE, MAE, MAPE, and directional accuracy, with Sharpe ratio included when the work is framed as a trading strategy.

Finally, **deployment** involves integrating the trained model into practical decision systems—such as algorithmic trading, risk management, or portfolio optimization—where the model is used for live or near-real-time inference.

Figure 3 illustrates the full forecasting workflow from problem definition through deployment monitoring. The pipeline explicitly includes baseline selection, backtesting, robustness checks, and post-deployment monitoring because these steps are central to evaluating whether reported model gains are reproducible and economically meaningful.

4 Results of the review and structured quantitative aggregation

4.1 Study selection and quantitative subset

The structured review process retained 103 studies for qualitative synthesis and identified 38 studies for the quantitative evidence map. Because these studies varied substantially in market domain, forecast horizon, data frequency, task formulation, and metric choice, they were not pooled into a single meta-analytic effect size. Instead, the quantitative evidence was coded into error-based forecasting, directional forecasting,

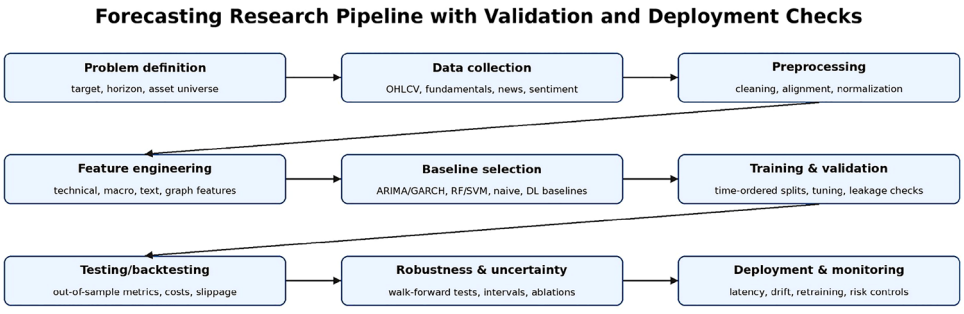


Fig. 3 Standard pipeline for stock market prediction research

trading-oriented evaluation, and robustness/uncertainty reporting. A focused paired-error aggregation was conducted for the directly recomputable subset.

The paired-error subset consists of 17 peer-reviewed studies and 47 same-dataset, same-horizon proposed-versus-baseline error comparisons. Model families represented in this subset include modular LSTM, attention-CNN/LSTM fusion, BiGRU-LSTM, PCA-ICA-LSTM, attention-enhanced BiLSTM, TCN-GRU-MHA, residual channel-spatial attention networks, DCGAN, VMD-TMFG-LSTM, optimized GRU/RNN/LSTM models, ES-LSTM/NARX variants, Transformer-versus-econometric comparisons, HDFM hierarchical decomposition, ICA-Multi-LSTM, RF-LSTM, Bi-LSTM, and SARIMA-LSTM. The broader quantitative evidence map includes directional, trading, and robustness evidence that is not pooled with point-forecast error metrics.

Accordingly, references to the quantitative evidence map in this review should not be interpreted as a pooled meta-analysis. Only studies with directly extractable same-task proposed-model and baseline error values were used to compute central tendencies. Other studies contributed directional, trading-oriented, domain-specific, robustness, or uncertainty evidence without being merged into the paired-error median. To make this filtering auditable, Appendix Tables A5 and A6 separate additional paired-error candidates from quantitative studies retained only for evidence mapping.

4.2 Error-based forecasting results

For studies with directly extractable paired error comparisons, relative error reduction was calculated between the proposed model and the strongest relevant external baseline reported within the same study. The paired-error subset includes 17 peer-reviewed studies and 47 paired comparisons. After first summarizing multiple pairs within each study by a study-level median, the cross-study median relative error reduction was 20.3%, with an interquartile range of 5.7%-50.7% and a full range from −0.8% to 71.5%. The cross-study median was 20.3% with all 17 studies and 16.8% after excluding the three largest study-level reductions of 71.5%, 67.3%, and 60.2%. These values should be interpreted as controlled, within-study comparisons rather than a universal estimate of ML/DL performance across all financial forecasting tasks. The inclusion of neutral, negative, and modest-gain study-level results reflects the conservative baseline rule and reduces cherry-picking risk (Table 1).

Several additional recent studies report quantitative outcomes that remain valuable for evidence mapping but are not merged into the paired-error median. For example, multimodal graph-sentiment, directional, volatility, and trading-oriented studies often report strong results, but their outputs may be expressed as maximum reductions, directional accuracy, Sharpe ratio, annualized return, or investment-value changes rather than directly comparable point-forecast errors. These studies are therefore retained for domain and robustness interpretation while being kept separate from the paired-error aggregation.

4.3 Directional forecasting and trading-oriented evidence

Directional forecasting studies frequently reported favorable results for deep learning, hybrid, feature-selection, and sentiment-aware models. However, many directional studies reported the winning model accuracy without a clean paired baseline accuracy suitable for percentage-point aggregation. These studies were therefore retained in the

Table 1 Directly recomputable paired-error evidence

Study	Domain	Metric/pairs	Proposed model vs baseline rule	Study-level median reduction	Aggregation note
Mostafavi & Hooman [17]	S&P 500 index	RMSE/1 pair	RF-LSTM vs S&P 500-only LSTM baseline	71.5%	Included; strong gain, but feature-set expansion is noted
Baek & Kim [18]	Equity indices	MAE ratio/1 pair	ModAugNet vs SingleNet	67.3%	Included; directly comparable paired-error study
Lin et al. [15]	Equities	Reported error/1 pair	STACN vs LSTM	60.2%	Included; multimodal text + price evidence
Osman and Otaibi [19]	S&P 500 index	RMSE/1 pair	Transformer vs ARIMA	54.8%	Included; daily S&P 500 data, 2015–2020
Sarıkoç and Çelik [20],	S&P 500 index	MAPE/1 pair	PCA-ICA-LSTM vs lean LSTM	46.5%	Included; dimension-reduction hybrid model
Zhang et al. [21]	Shanghai equities	RMSE/5 pairs	VMD-TMFG-LSTM vs strongest external baseline	43.4%	Included; study-level median across five stocks
He & Dai [22]	Chinese securities equities	Network RMSE/1 aggregate pair	ICA-Multi-LSTM vs Multi-LSTM	38.8%	Included; network-level test M-RMSE used to avoid cherry-picking single stocks
Ghosh et al. [23]	Equities/indices	RMSE/5 pairs	TCN-GRU-MHA vs random-walk baseline	24.4%	Included; 1-day horizon values used
Sezgin et al. [24]	Airline equities	RMSE/2 pairs	Optimized GRU vs strongest reported LSTM comparator	20.3%	Included; values transcribed from published table.
Conti [25],	FTSE MIB stocks	RMSE/10 pairs	DCGAN vs strongest non-proposed baseline per stock	20.0%	Included; study-level median across ten stocks
Liu [26]	Large-cap equities	RMSE/4 pairs	M-A-BiLSTM vs strongest external benchmark	13.6%	Included; four stock datasets
Shaban et al. [27]	IBM equity	RMSE/1 pair	SMP-DL vs strongest reported non-proposed comparator	9.0%	Included; conservative comparator replaces weaker baseline
Li et al. [28]	SSEC/SZI/SPX stock indices	RMSE/3 pairs	HDFM vs strongest external comparator (McVCsB)	7.2%	Included; study-level median across three indices
Sun et al. [29]	US/China equities	RMSE/4 pairs	RCSAN vs strongest external comparator	4.1%	Included; modest gain against strong baselines
Seabe et al. [30]	Cryptocurrencies	RMSE/3 pairs	Bi-LSTM vs strongest LSTM/GRU comparator	1.2%	Included; neural-vs-neural crypto comparison retained with modest median gain
Adesina & Obokoh [31]	FX (ZAR/USD)	RMSE/1 pair	SARIMA-LSTM vs strongest reported LSTM comparator	0.03%	Included; MAPE excluded because several FX MAPE values are infinite/unstable
Cocianu et al. [32]	Crypto/FX	RMSE/3 pairs	ES-LSTM/NARX vs standard LSTM	−0.8%	Included; neutral-to-negative result retained

Relative error reduction was computed as [(baseline error - proposed model error)/baseline error] x 100. When a study reported multiple same-dataset, same-horizon comparisons, the table reports the study-level median reduction before cross-study aggregation

directional evidence map but not pooled into a single median directional gain. Trading-oriented studies were analyzed separately because profitability depends on execution assumptions, transaction costs, slippage, liquidity, risk exposure, and market regime.

Trading-oriented outcomes in Table 2 are reported as source-study results rather than as evidence of deployable profitability. Extreme backtest returns should be interpreted cautiously because they may depend on market period, leverage assumptions, omitted transaction costs, slippage, liquidity constraints, exchange conditions, and strategy-specific reward design.

4.4 Robustness, uncertainty, and reporting limitations

The quantitative evidence is stronger for reported predictive improvements than for real-world robustness. Most extractable studies emphasized point forecast metrics such as RMSE, MSE, MAE, MAPE, accuracy, or Sharpe ratio. By contrast, calibrated prediction intervals, interval widths, empirical coverage, and uncertainty reliability were rarely available in a form that could be aggregated across the quantitative evidence map. For this reason, uncertainty quantification is treated as an underreported methodological gap rather than as a mature quantitative conclusion supported across the reviewed literature. No reliable median prediction-interval width, empirical coverage rate, or calibration statistic could be computed from the evidence map (Table 3).

These findings reconcile the positive performance results with the limitations identified later in the review. Modern ML/DL models often improve reported forecasting metrics under controlled experimental conditions, but the evidence does not establish universal real-world superiority across assets, horizons, and market regimes. Statistical gains in historical experiments should therefore be interpreted alongside validation quality, baseline strength, data leakage controls, and market-friction assumptions.

5 Evolution of models for stock market forecasting

The evolution of financial forecasting models has moved from rigid linear assumptions toward flexible, data-driven architectures capable of capturing the market's inherent nonlinearity and volatility.

Table 2 Directional, sentiment, and trading-oriented evidence

Study	Domain	Main model or comparison	Reported quantitative outcome	Interpretation caveat
Brauneis & Sahiner [33]	Cryptocurrency volatility	LightGBM, XGBoost, LSTM with sentiment vs no-sentiment settings	Sentiment-enhanced ML forecasts improved no-sentiment forecasts in 54.17% of studied cases	Evidence supports nonlinear sentiment effects, not a universal sentiment benefit
Omole & Enke [34]	Bitcoin direction/trading	Boruta + CNN-LSTM and backtested trading strategies	82.44% directional accuracy; one long-short strategy reported 6654% annual return	Backtest return is highly sensitive to strategy design and market frictions
Sattarov & Choi [35]	Bitcoin trading	Multi-level DQN with Twitter sentiment	29.93% increase in investment value; Sharpe ratio above 2.7	Economic result depends on reward design, market period, and execution assumptions
Yang & Malik [36]	Cryptocurrency pair trading	RL pair trading vs traditional pair trading	RL annualized profits of 9.94%-31.53% vs 8.33% for traditional pair trading	Trading outcome should be interpreted with transaction-cost, slippage, and regime caveats

Table 3 Robustness and uncertainty reporting patterns across the quantitative evidence map

Evidence dimension	Observed reporting pattern	Aggregation status	Implication for this review
Point forecast error	Often reported using RMSE, MSE, MAE, or MAPE	Directly aggregated when paired model-baseline values were extractable	Supports conservative paired error comparisons where baselines are extractable
Directional accuracy	Often reported, but paired baselines are not always extractable	Mapped; pooled only when paired baseline accuracies are available	Useful for evidence mapping; limited pooled percentage-point aggregation
Trading performance	Reported using return, investment value, annualized profit, or Sharpe ratio	Summarized separately from forecast-error metrics	Kept separate because results depend on costs, slippage, liquidity, and risk
Uncertainty intervals	Prediction intervals, interval widths, coverage, and calibration are reported inconsistently and often absent	Not pooled	No reliable aggregation of interval width or coverage was possible
Cross-regime robustness	Walk-forward validation, cross-period testing, and regime-shift analysis are unevenly reported	Used as quality flags	Limits claims of stable out-of-sample superiority

5.1 Classical statistical and early machine learning models

- **Statistical Foundations:** Early forecasting relied on models like ARIMA and GARCH. However, these rely on assumptions of linearity and stationarity that frequently fail in dynamic market conditions.
- **Support Vector Machines (SVM):** SVMs were among the first ML tools used to simulate intricate dependencies in financial time series [5].
- **Random Forests:** Tree-based ensemble methods provided a nonlinear alternative to classical econometrics [6].
- **Linear-Nonlinear Hybrids:** To address both patterns, researchers developed hybrid models such as the ARIMA-LSTM architecture.

5.2 Recurrent neural networks and LSTM-Based models

- **Sequential Modeling:** RNNs were introduced to model the temporal dependencies intrinsic to financial time series.
- **Long Short-Term Memory (LSTM):** LSTMs addressed the "vanishing gradient" problem of standard RNNs, allowing the model to remember long-range market dependencies [37].
- **Gated Recurrent Units (GRU):** GRUs emerged as an efficient alternative to LSTMs, and some recent cryptocurrency and high-frequency studies report strong GRU performance under specific data and validation regimes [38].

5.3 CNN-Based and hybrid architectures

- **Feature Extraction:** Convolutional Neural Networks (CNNs) are used to extract local features and "visual" patterns from market data [39].
- **CNNpred:** Specialized CNN architectures were developed to predict trends by integrating diverse sets of variables [40, 41].
- **Hybrid CNN-LSTM:** This architecture combines CNN layers for local feature extraction with LSTM layers for temporal dependencies, often used in limit order book forecasting.

5.4 Transformer and attention-based models

- **Self-Attention Mechanisms:** Transformers process entire sequences in parallel and can reduce some limitations of recurrent models, especially for longer context windows [42]. In finance, however, Transformer superiority remains conditional on data regime, feature design, baseline strength, and validation protocol rather than universal.
- **Time-Series Specialization:** Models like the Informer [43] and Autoformer [44] utilize auto-correlation and decomposition components for long-term forecasting.
- **Lightweight Variants:** For real-time scalability, variants such as Performer [45] and Linformer [46] employ kernel approximations to reduce computational complexity.
- **Foundation Models for Time Series:** Recent (2023–2024) work has extended Transformer-based forecasting toward universal foundation models. PatchTST [47] introduced patch-based input tokenization for long-horizon forecasting. TimeGPT [48] and Chronos [49] demonstrate that LLM-style pretraining on time-series corpora enables zero-shot forecasting on financial signals, while Moirai [50] and TimesFM [51] extend this to multi-domain and probabilistic settings. While benchmarking on financial markets remains preliminary, early results suggest competitive performance against domain-specific baselines without per-asset retraining.

5.5 Multimodal, ensemble, and reinforcement learning systems

- **Multimodal Fusion:** Modern systems integrate price data with unstructured information, such as social media sentiment and news [52].
- **Domain-Adapted NLP:** Models like FinBERT are utilized to extract signals from earnings calls and financial filings [53].
- **Advanced Ensembles:** Models such as XGBoost and LightGBM are frequently combined with deep learning to capture nonlinear market influences.
- **Reinforcement Learning (RL):** RL systems, including Deep Q-Networks (DQN), are used to directly optimize trading policies rather than just price direction [54].

6 Applications by market domain

This section compares how market structure affects data availability, target construction, model choice, validation design, and generalizability across financial domains. Model architectures and quantitative evidence are discussed in Sects. 4 and 5, while broader limitations and research opportunities are addressed in Sects. 7 and 8.

6.1 Equity indices

Studies focusing on equity indices typically frame prediction tasks around index-level returns, directional movements, or volatility, often at daily or multi-day horizons. Because indices aggregate multiple constituent assets, short-term price fluctuations tend to be smoother than those observed in individual securities, which influences both target selection and evaluation sensitivity.

Index-based applications are often shaped by macroeconomic exposure and cross-sector aggregation, with some studies incorporating macroeconomic indicators or

market-wide sentiment proxies. As a result, reported outcomes usually emphasize directional accuracy or risk-related metrics rather than precise point forecasts.

Performance results in this domain are commonly reported using predictive accuracy measures or portfolio-level metrics derived from index tracking or allocation scenarios. These results are generally presented within the context of a specific index and time period, reflecting domain-specific characteristics rather than general model behavior.

6.2 Individual equities

Applications targeting individual equities span a wide range of experimental settings, including single-stock prediction, multi-stock modeling, and sector-based grouping approaches. Prediction horizons vary from short-term directional movement to longer-term return estimation, depending on the intended application.

A key constraint in individual equity studies is event sensitivity, as earnings announcements, firm-specific news, and corporate actions can induce abrupt price movements. This sensitivity influences both model inputs and evaluation outcomes, particularly in short-horizon prediction tasks.

At a high level, feature selection in this domain often reflects the need to balance firm-specific information with broader market signals. While implementation details differ across studies, individual equity applications typically emphasize adaptability to heterogeneous asset behavior rather than uniform modeling assumptions.

6.3 Commodities and FX

Machine learning applications in commodities and foreign exchange markets are commonly motivated by pricing, directional forecasting, or risk and volatility estimation. Compared to equity markets, these domains are often influenced by exogenous drivers, including supply constraints, geopolitical developments, and monetary policy decisions.

Such external factors introduce domain-specific constraints that affect model stability and generalization. As a result, studies frequently focus on limited asset subsets or specific economic regimes, which shapes both experimental design and reported outcomes.

Evidence from these markets suggests that transferability across assets or time periods is often limited, with models tuned to particular instruments or conditions. Consequently, reported findings are typically framed as context-dependent rather than broadly generalizable.

6.4 Cryptocurrencies

Cryptocurrency markets represent a distinct application regime within the financial forecasting literature because they differ from traditional equity markets in trading hours, market maturity, exchange fragmentation, sentiment sensitivity, liquidity structure, and regulatory uncertainty. These features make cryptocurrency studies useful for testing adaptive and sentiment-aware models, but they also make cross-period generalization especially fragile.

6.4.1 Distinct market structure

Cryptocurrency trading occurs on a continuous, 24/7 basis across multiple exchanges, leading to differences in liquidity dynamics, market microstructure, and price formation.

Exchange fragmentation and varying market rules further contribute to heterogeneity in observed data.

6.4.2 Common tasks and reporting styles

Most cryptocurrency-focused studies examine short-horizon directional prediction, volatility forecasting, or price movement classification, often motivated by high market volatility. Some works frame predictions within trading-oriented evaluations, while others report purely predictive metrics.

Reported results are typically presented within narrow experimental settings, with performance evaluated over specific assets, time periods, or market conditions.

6.4.3 Data regime characteristics

The data regimes used in cryptocurrency studies vary widely, ranging from daily price series to intraday data and, less commonly, order book–level observations. These choices reflect study-specific objectives and data availability rather than standardized preprocessing conventions.

6.4.4 Evidence reporting caveats

A recurring characteristic of cryptocurrency applications is the limited and inconsistent treatment of transaction costs, liquidity constraints, slippage, exchange fragmentation, and regime instability. Extreme backtest results can therefore be informative as proof-of-concept evidence, but they should not be interpreted as stable deployable profitability unless cost-aware, walk-forward, and cross-regime tests are reported.

Cryptocurrency findings are therefore best interpreted by task type. Sentiment-based volatility studies suggest that news and social signals can add value when market attention drives rapid price moves, while directional and trading-oriented studies show that CNN-LSTM and reinforcement-learning systems can capture short-horizon nonlinear patterns. However, accuracy or backtested return gains are not directly comparable across assets, exchanges, sampling frequencies, horizons, and cost assumptions, so these results are treated as domain-specific evidence rather than pooled proof of deployable profitability (Table 4).

Across domains, model performance depends less on architecture alone than on whether the validation design matches the market structure being forecasted. Index studies require macro-regime awareness, individual-equity studies require event and survivorship controls, commodities and FX studies require sensitivity to exogenous shocks, and cryptocurrency studies require explicit treatment of exchange fragmentation, liquidity, slippage, transaction costs, and 24/7 regime instability.

7 Challenges and limitations

This section synthesizes recurring limitations reported across the reviewed literature and explains factors that can reduce the stability or real-world relevance of reported performance. Proposed methodological solutions and emerging research directions are discussed separately in Sect. 8.

Table 4 Domain-specific forecasting constraints and evaluation implications

Market domain	Common targets	Useful signals	Typical model/evaluation issue	Main generalization risk
Equity indices	Returns, direction, volatility	OHLCV, macro indicators, sector sentiment, index constituents	Smoother aggregate targets, but results depend on regime-aware validation and strong baselines	Macro regime shifts and sector rotation can weaken historical relationships
Individual equities	Firm-level returns, price direction, event response	Earnings, firm news, fundamentals, technical indicators, market-wide features	High event sensitivity requires careful feature timing and survivorship controls	Corporate events, delistings, and firm-specific shocks reduce transferability
Commodities and FX	Prices, volatility, direction, risk exposure	Rates, supply-demand variables, macro releases, geopolitical and policy signals	Exogenous shocks can dominate learned price patterns, making limited-window validation fragile	Models tuned to one instrument or regime may transfer poorly across assets or periods
Cryptocurrencies	Direction, volatility, trading policy, short-horizon price movement	Exchange prices, order books, liquidity, social sentiment, news, on-chain or market activity signals	Continuous trading, exchange fragmentation, and unstable liquidity complicate backtesting and execution assumptions	Slippage, market fragmentation, regulatory shocks, and rapid regime shifts limit deployable stability

7.1 Data validity and bias

Several forms of data-related bias are discussed across prior studies, including risks of information leakage and look-ahead effects that can arise when future information is unintentionally incorporated into training or feature construction. Such issues can inflate apparent predictive performance and complicate interpretation of reported results.

Additional concerns relate to survivorship and selection bias, particularly in equity-focused datasets where delisted or inactive assets may be excluded. These exclusions can alter the statistical properties of the sample and affect the generality of reported findings.

Regime shifts and structural breaks represent a further validity challenge. Because financial markets evolve over time, relationships observed in one period may weaken or reverse in another, limiting the temporal stability of learned patterns.

7.2 Evaluation realism and comparability

Evaluation procedures vary substantially across studies, including differences in train–test split strategies, forecasting horizons, retraining frequency, and normalization choices. These variations influence reported outcomes and make direct comparison across papers difficult.

Reported metrics also differ widely depending on task formulation. Some studies emphasize statistical error measures, while others report directional accuracy or portfolio-level financial indicators. Because these metrics capture different objectives, improvements in one measure do not necessarily translate into meaningful gains in another.

A related distinction arises between statistical significance and economic relevance. Even when predictive improvements are measurable, their practical importance may depend on transaction costs, execution assumptions, and risk exposure, which are not always consistently incorporated.

7.3 Robustness and generalization

Many studies note sensitivity of model performance to hyperparameter configuration, training window selection, and evaluation period. Such sensitivity can indicate overfitting to specific historical conditions rather than stable predictive structure.

Generalization across assets, markets, or time periods is also reported inconsistently. Models that appear effective in one dataset or regime may not retain similar behavior elsewhere, highlighting variability in empirical outcomes rather than uniform performance patterns.

Reproducibility presents an additional concern, as implementation details, preprocessing steps, or ablation analyses are sometimes only partially documented. Limited transparency can make independent verification challenging and contributes to uncertainty in interpreting reported results.

7.4 Interpretability and uncertainty reporting

Across the reviewed literature, interpretability and uncertainty estimation are often discussed only briefly or omitted entirely. Reported predictions may lack clear explanations of contributing factors or quantified confidence levels, which can restrict their usefulness in risk-sensitive contexts.

The absence of transparent uncertainty characterization can also complicate comparison between studies, since predictive reliability may differ even when point estimates appear similar. These limitations highlight the importance of interpretability and uncertainty awareness for trustworthy financial modeling, though specific methodological responses are considered later in Sect. 8.

7.5 Deployment constraints in practice

Practical deployment considerations introduce further challenges beyond experimental evaluation. Computational latency, infrastructure requirements, and real-time data availability can influence whether reported modeling approaches are operationally feasible.

Data pipeline stability and monitoring also emerge as recurring concerns. Changes in data distribution, vendor definitions, or market structure may degrade performance over time, requiring ongoing validation and maintenance. These operational factors illustrate how real-world constraints can differ substantially from controlled experimental settings.

8 Future research directions in AI-driven stock market forecasting

The continued advancement of machine learning in finance requires methodological improvements that address current limitations before new predictive capabilities can be trusted in practice. Several research areas are likely to influence the next stage of financial forecasting research (Fig. 4).

8.1 Scalability and real-time processing

High-frequency trading (HFT) creates demanding requirements for computational speed, data latency, and model efficiency. To address these constraints, researchers have explored lightweight Transformer variants such as Performer and Linformer, which use kernel approximations or low-rank factorization to reduce attention complexity [45, 46].

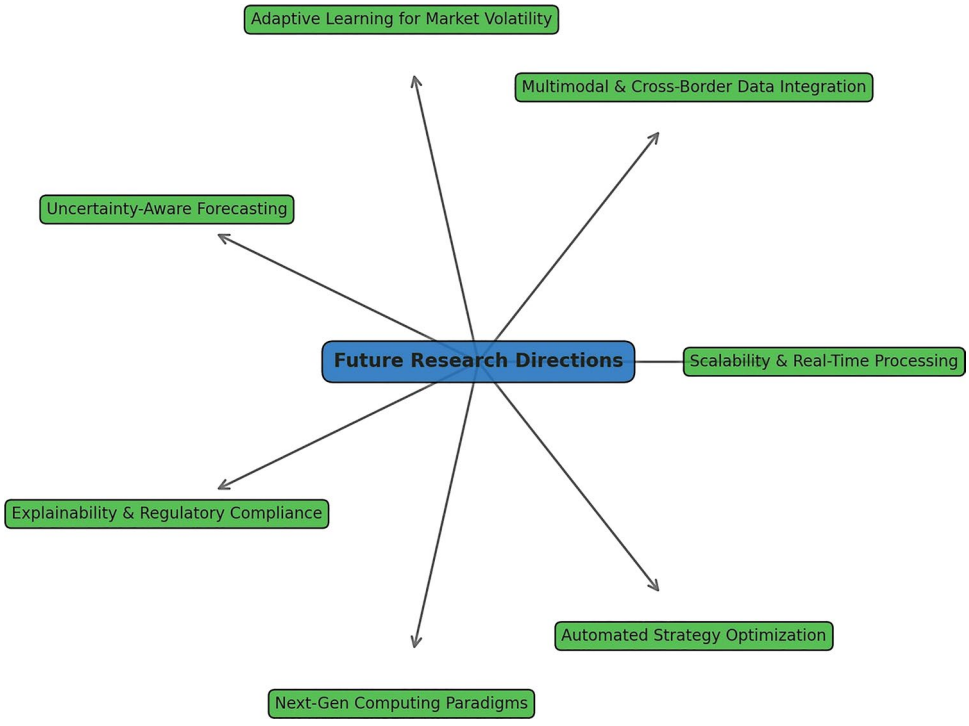


Fig. 4 Future research directions in data science and machine learning

Model compression techniques, including pruning [55] and quantization [56], along with specialized hardware accelerators [57], may also support more efficient inference. These approaches are relevant to real-time forecasting, although their practical value depends on data-feed stability, execution latency, and monitoring after deployment.

8.2 Multimodal and cross-border data integration

Effective financial forecasting increasingly relies on synthesizing diverse data types. Cross-modal models integrate structured numerical market data with unstructured textual information from sources such as social media sentiment [58], financial news reports [52], and earnings call transcripts [59]. Multilingual models and translation-enabled pipelines may also support analysis of international news sources and central bank announcements, which can be relevant for global indices and foreign exchange markets [60]. The development and curation of robust multimodal and multilingual financial datasets remain important because model gains are highly sensitive to data timing, source quality, and leakage control.

8.3 Adaptive learning for market volatility

Financial markets exhibit dynamic, non-stationary behavior, requiring forecasting models capable of adaptive learning. Meta-learning frameworks enable models to rapidly adapt to market regime shifts [61], and online learning algorithms facilitate continuous parameter adjustments [62]. Transfer learning techniques can also bridge domain disparities, such as applying insights from established markets to less liquid or emerging markets [63]. To effectively evaluate these approaches, novel evaluation methods emphasizing model performance during volatile periods are essential [64].

8.4 Uncertainty-aware forecasting

Accurate financial forecasting necessitates moving beyond point predictions to incorporate uncertainty quantification. Bayesian deep learning methods, such as Monte Carlo dropout [65] and variational inference [66] integrated with LSTM networks [37], generate probabilistic predictions and confidence intervals. Evidential neural networks offer further potential for directly modeling uncertainty distributions [67]. Complementary visual analytics tools are needed to effectively communicate these insights to financial decision-makers [68].

8.5 Explainability and regulatory compliance

Growing regulatory scrutiny necessitates explainable and transparent forecasting models. Techniques such as layer-wise relevance propagation [69] and integrated gradients [70] enhance interpretability. Surrogate modeling approaches, such as symbolic regression, create simplified, auditable representations of complex neural networks [71]. Establishing standardized benchmarks for explainability and creating regulatory "sandboxes" are essential for broad institutional acceptance, particularly within frameworks such as the European Union's AI Act [72].

8.6 Next-generation computing paradigms

Next-generation computing paradigms may provide future opportunities, but their current role in financial forecasting remains exploratory. Quantum machine learning has been proposed for complex optimization problems [73], while heterogeneous computing can support larger-scale model training and inference [74]. Adaptive signal processing methods, including dynamic mode decomposition [75] and learned wavelet transforms [76], may also improve noise reduction and feature extraction, although their forecasting value should be tested under realistic validation protocols.

8.7 Automated strategy optimization

Combining automated machine learning (AutoML) with financial performance optimization is an active research direction rather than a settled solution. Techniques such as neural architecture search and meta-learning can help identify candidate forecasting architectures [77], while reinforcement learning methods directly optimize trading policies rather than only point forecasts [54]. These approaches require rigorous safeguards against overfitting, data leakage, and unrealistic backtest assumptions [78].

Collectively, these research trajectories suggest a future in which adaptive, explainable, and uncertainty-aware systems integrate multimodal and cross-market data in near real time. The evidence reviewed here suggests that these directions are promising, but their value will depend on stronger benchmarks, realistic market-friction modeling, transparent reporting, and validation across changing market regimes.

9 Conclusion

This review finds that machine learning and deep learning methods have substantially expanded the methodological toolkit for financial forecasting, but their reported gains remain conditional. The paired-error synthesis included 17 peer-reviewed studies and 47 same-dataset, same-horizon proposed-versus-baseline comparisons suitable for direct error-based aggregation. The study-level median relative error reduction was 20.3%, with an

interquartile range of 5.7%-50.7% and a full range from −0.8% to 71.5%. These values indicate frequent improvement in controlled comparisons, not universal real-world superiority.

The evidence base remains uneven. Some models reported large reductions relative to their strongest available baselines, while others produced modest or negative study-level improvements. Reported gains vary across market domains, data frequencies, prediction horizons, model families, feature regimes, validation protocols, and market regimes. In addition, many studies provide limited evidence on cross-regime robustness, transaction-cost-aware profitability, slippage, calibrated uncertainty estimates, or reproducible validation procedures.

The main conclusion is therefore conditional rather than absolute. ML/DL models often improve reported forecasting performance under controlled experimental settings, but confidence in their external validity is constrained by heterogeneous tasks and uneven methodological rigor. Future work should prioritize rolling and cross-regime validation, stronger baseline design, realistic market-friction modeling, reproducibility, explainability, and uncertainty-aware forecasting. These steps are necessary for moving financial forecasting research beyond isolated accuracy gains toward more reliable decision-support systems.

Appendix

A. Extended tables

Table A1 Datasets frequently used in forecasting research

Dataset/Source	Frequency	Key Features	Applications
Yahoo Finance/Bloomberg	Daily, Intraday	OHLC prices, volume	General stock & index forecasting
Tushare	Minute-level	Chinese market focus	Emerging markets, high-frequency prediction
CRSP/Compustat	Daily	U.S. equities, financial fundamental	Firm-level forecasting
Kaggle Stock Datasets	Variable	Curated challenges, multiple markets	Benchmarking new models
Crypto Exchanges (Binance, Coinbase)	Tick-level	Price, order book, sentiment streams	Cryptocurrency forecasting

Table A2 Preprocessing techniques and their roles

Technique	Purpose	References
Wavelet Transform	Noise reduction, multi-resolution decomposition	Wavelet Transform
Empirical Mode Decomposition (EMD)	Adaptive signal decomposition	Huang et al. [79]
Min-Max Scaling	Normalization to [0,1]	Ioffe & Szegedy [80]
Z-score Normalization	Standardization, handling outliers	Fischer & Krauss [81]
Sentiment Lexicons (Loughran-McDonald)	Extracting textual features	Loughran & McDonald [59]

Table A3 Hybrid and ensemble architectures

Model Type	Composition	Application
CNN-LSTM	CNN layers for local feature extraction, LSTM for temporal dependencies	CNN-LSTM
ARIMA-LSTM	ARIMA for linear trends, LSTM for nonlinear residuals	ARIMA-LSTM
Wavelet-LSTM	Decompose signals before LSTM training	Wavelet-LSTM
Multimodal LSTM-BERT	Combine text sentiment and price data	Event-driven stock forecasting
Ensemble Meta-Learners	Stacking CNN/RNN/Transformer outputs	Volatile market forecasting

B. Quantitative evidence audit trail

Table A4 records the paired-error audit trail used for the central-tendency results. The 38-study quantitative evidence map informed the structured synthesis, while the directly recomputable paired-error subset supplied 17 peer-reviewed studies and 47 same-dataset, same-horizon paired comparisons. Only these studies were used for the central tendency estimates in Sect. 4.2. Directional, trading-oriented, robustness, and uncertainty evidence remains mapped separately because those outcomes are not commensurable with RMSE, MSE, MAE, or MAPE. Tables A and A preserve the screening trail by separating additional paired-error candidates from the 21 quantitative evidence-map studies retained outside the aggregation.

Table A4 Audit trail for directly recomputable paired-error evidence

Study	Pairs/metric	Baseline selection	Study-level median	Use/caveat
Mostafavi & Hooman [17]	1/RMSE	S&P 500-only LSTM baseline	71.5%	Included; feature-set expansion noted
Baek & Kim [18]	1/MAE ratio	SingleNet baseline	67.3%	Paired-error aggregation
Lin et al. [15]	1/error value	LSTM baseline	60.2%	Paired-error aggregation
Osman & Otaibi [19]	1/RMSE	ARIMA econometric baseline	54.8%	Included; S&P 500 daily data, 2015–2020
Sarıkoç and Çelik [20],	1/MAPE	Lean LSTM baseline	46.5%	Paired-error aggregation
Zhang et al. [21]	5/RMSE	Strongest external baseline across ARIMA/NN/DNN/CNN/LSTM	43.4%	Paired-error aggregation
He & Dai [22]	1/network M-RMSE	Multi-LSTM aggregate network benchmark	38.8%	Included; network-level comparison retained instead of selecting single-stock rows
Ghosh et al. [23]	5/RMSE	Random-walk baseline for 1-day horizon	24.4%	Paired-error aggregation
Sezgin et al. [24]	2/RMSE	Strongest reported non-GRU comparator	20.3%	Values transcribed from published table.
Conti [25],	10/RMSE	Strongest non-proposed model per stock	20.0%	Paired-error aggregation
Liu [26]	4/RMSE	Strongest external benchmark among MLP/LSTM/GRU/CNN/BiLSTM/CNN-LSTM	13.6%	Paired-error aggregation
Shaban et al. [27]	1/RMSE	Strongest reported non-proposed comparator	9.0%	Paired-error aggregation
Li et al. [28]	3/RMSE	Strongest external comparator, McVCsB, for SSEC/SZI/SPX	7.2%	Included; median across three stock indices
Sun et al. [29]	4/RMSE	Strongest external comparator for each dataset	4.1%	Paired-error aggregation
Seabe et al. [30]	3/RMSE	Bi-LSTM vs strongest LSTM/GRU comparator per crypto asset	1.2%	Included; crypto neural-baseline comparison
Adesina & Obokoh [31]	1/RMSE	SARIMA-LSTM vs strongest reported LSTM comparator for ZAR/USD	0.03%	Included conservatively; MAPE excluded
Cocianu et al. [32]	3/RMSE	Standard LSTM benchmark	−0.8%	Negative result retained

Table A5 Additional quantitative studies considered for paired-error aggregation but not included in the central estimates

Study	Domain/data	Candidate paired comparison	Extractable metrics or values	Aggregation decision	Main caveat
Mostafavi & Hooman [82]	S&P 500 daily forecasting with 88 technical indicators	LSTM, XGBoost, RF, and SVR comparisons	Reported MAE, MAPE, and RMSE for lagged and non-lagged settings	Not aggregated	Not aggregated because the LSTM error scale was not directly commensurable with the tree/SVR rows.
Kulkarni et al. [83]	BSE stock data, pre-COVID and COVID periods	ARIMA, SMA, Holt-Winters, SVR, RFR, XGBoost, LSTM, and MLP	Reports RMSE, MAE, and MAPE by model and period	Not aggregated	Not aggregated because the full same-task paired baseline values were not consistently extractable from the available reporting.
Pan [84]	Bitcoin, Ether, and Dogecoin, 2018–2022	LSTM vs Random Forest vs ARIMA	Reports MSE, RMSE, MAE, and R2	Not aggregated	Not aggregated because venue/reporting limitations did not support unambiguous paired-error extraction.
Chen [85]	Bitcoin daily data, 2015–2022	RF and LSTM with DM/CW tests	Reports MAPE, RMSE, and directional accuracy	Not aggregated	Relevant crypto comparison retained in the evidence map, but not aggregated because exact same-task paired values were not consistently extractable.
Lin et al. (2021)	S&P 500 and CSI 300	CEEMDAN-LSTM vs single and mixture models	Uses forecast-error criteria and MCS testing	Not aggregated	Not aggregated because exact paired values were not consistently extractable from the reported text and tables.
Pilla & Mekonen [86]	S&P 500 daily data	LSTM vs ARIMA	Reports ARIMA and LSTM RMSE/MAE values	Not aggregated	Retained only as contextual evidence because of preprint status; excluded from the peer-reviewed paired-error aggregate.

These studies were retained as a screening trail but were not merged into the 17-study paired-error aggregate. They may support future synthesis if raw values, scaling, baseline equivalence, and horizon comparability can be verified without ambiguity

Table A6 Quantitative evidence-map studies retained outside the 17-study paired-error aggregation

Study	Domain/task	Quantitative evidence type	Reason not pooled with Table A4	Evidence-map use
Akita et al. [87]	Individual equities with numerical and textual information	Directional/classification-style stock prediction	Not a same-horizon paired point-error comparison	Multimodal price + text evidence
Fischer & Krauss [81]	S&P 500 constituents	Return prediction/trading-oriented evaluation	Trading and classification framing is not commensurable with RMSE/MAE/MAPE aggregation	Early LSTM equity evidence
Huang, Nakamori & Wang [5]	Stock index movement direction	SVM directional forecasting	Directional accuracy rather than point forecast error	Classical ML baseline context
Hoseinzadeh & Haratizadeh [40, 41]	Multiple stock markets	CNN-based directional market prediction	Classification output and heterogeneous market settings	CNN evidence in market prediction
Zhang, Zohren & Roberts [88]	Limit order books	DeepLOB classification/microstructure prediction	Order-book classification task differs from daily price-error forecasting	High-frequency domain evidence

Table A6 (Continued)

Study	Domain/task	Quantitative evidence type	Reason not pooled with Table A4	Evidence-map use
Kim, Choi & Lee [89]	Equity price prediction	Attention-CNN quantitative forecasting	Not used because comparable source-level paired baseline extraction was not completed	Attention/CNN evidence
Nguyen & Yoon [90]	Financial time-series forecasting	Attention-based BiLSTM metrics	Not aggregated because exact same-task external baseline pairing was not sufficiently clear for aggregation	Attention-BiLSTM evidence
Xu, Zhang & Li [91]	News + price multimodal forecasting	Multimodal price/news metrics	Metric and baseline structure not directly commensurable with paired-error subset	Text-price fusion evidence
Wang, Chen & Wang [92]	Stock price prediction	Dilated CNN quantitative metrics	Not pooled without same-dataset external baseline extraction	CNN-family evidence
Wang, Xu & Chen [93]	Wavelet hybrid stock forecasting	Hybrid/decomposition forecasting metrics	Baseline and scaling comparability not source-verified for paired aggregation	Decomposition-hybrid evidence
Nguyen, Nguyen & Ngo [16]	Multimodal/graph-sentiment stock forecasting	Forecasting metrics with sentiment fusion	Multimodal setup not directly comparable to point-error-only aggregate	Recent graph/sentiment evidence
Brauneis & Sahiner [33]	Cryptocurrency volatility	Sentiment and ML volatility forecasting	Volatility/sentiment results not same as price-error pair	Crypto volatility evidence
Omole & Enke [34]	Bitcoin direction and trading	Directional accuracy and backtested returns	Directional/trading metrics are not RMSE/MAE/MAPE	Trading-oriented crypto evidence
Sattarov & Choi [35]	Bitcoin trading	Investment value and Sharpe ratio	Reinforcement-learning trading outcome, not point forecast error	RL trading evidence
Yang & Malik [36]	Cryptocurrency pair trading	Annualized profit and pair-trading return metrics	Trading profits depend on costs, slippage, and strategy design	Crypto RL/pair-trading evidence
Foroutan & Lahmiri [94]	Crude-oil price forecasting	Deep learning/econometric commodity forecasts	Commodity-specific metrics not source-paired into Table A4	Commodity-domain evidence
Rodrigues & Machado [38]	High-frequency cryptocurrency forecasting	Deep learning price/return metrics	High-frequency crypto setting separated from daily stock-error median	High-frequency crypto evidence
Jagadesh et al. [14]	Stock trend prediction using Heiken-Ashi candles	Trend-classification/ML metrics	Trend labels are not directly commensurable with point forecast errors	Candlestick-feature evidence
Huang, Capretz & Ho [6]	Stock market prediction	Multi-objective ensemble learning metrics	Not used without exact paired same-horizon error extraction	Tree/ensemble evidence
Patel & Yalamanchili [95]	Stock prediction	Ensemble comparison metrics	Comparative evidence retained, but not a verified paired-error row	Ensemble-learning evidence
Ramos-Pérez et al. [96]	Stock prediction with Transformers/attention	Transformer/attention quantitative results	Metric formulation differs from paired point-error subset	Transformer evidence

This table closes the audit gap between the 38-study quantitative evidence map and the 17-study paired-error aggregation. These 21 studies provide directional, trading-oriented, domain-specific, robustness, uncertainty, or otherwise quantitative evidence, but they were not pooled into the paired-error median because their outcomes are not directly commensurable with same-dataset, same-horizon RMSE, MSE, MAE, or MAPE pairs

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Consent to participate
Not applicable.

Consent for publication
NA.

Accordance statement
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Competing interests
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